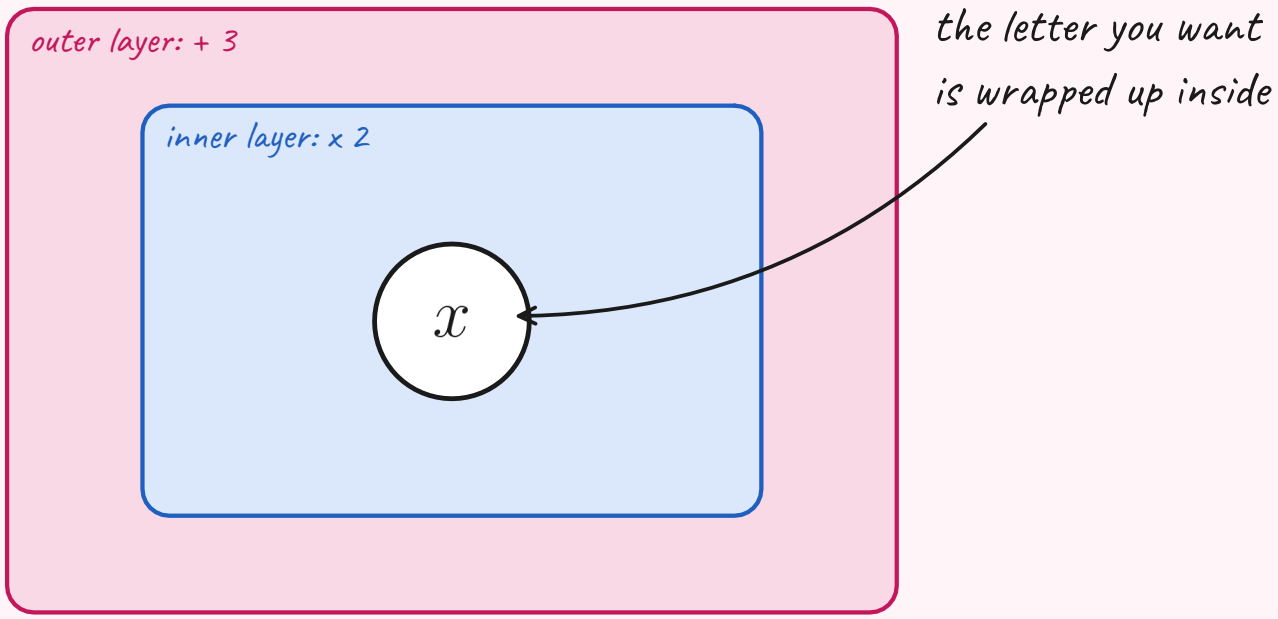
Chapter 14

Transposing formulae

or: unwrapping a present to get at the letter inside



A formula is usually written to work out **one** particular thing. $F = \frac{9}{5}C + 32$ turns a temperature in Celsius into Fahrenheit. But what if you know the Fahrenheit and want the Celsius? You need to **rearrange** the formula so that C is on its own.

Rearranging a formula is called **transposing** it. The trick in this chapter is to see every formula as a present: the letter you want is in the middle, and the formula has wrapped it in layers ("times 2", "add 3", "square root"...). To get the letter out, you take the layers off, **outside layer first**.

Goes with **Lecture 2** (Algebraic manipulation) and parts of Lectures 7 and 12. Uses: letters and substituting (Chapter 8), squares and roots (Chapter 9), brackets (Chapter 11), factorising (Chapter 12) and algebraic fractions (Chapter 13).

What "make x the subject" means

The **subject** of a formula is the letter that is **on its own** on one side of the = sign. In

$$y = 2x + 3$$

the subject is y. The formula is a machine for working out y: give it an x and it tells you y. **"Make x the subject"** means: rearrange it so it looks like

$$x = (\text{something with } y \text{ in it})$$

so that it becomes a machine for working out x instead. The words **transpose**, **rearrange**, **change the subject**, **make x the subject** and **solve for x** all mean the same thing.

The undo buttons

RULE: inverse operations (undo buttons)

$+$ is undone by $-$, and $-$ by $+$.

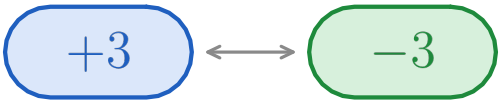
$\times$ is undone by $\div$, and $\div$ by $\times$ (**never divide by something that could be 0**).

Squaring is undone by square-rooting (giving $\pm$), and $\sqrt{}$ by squaring.

Cubing is undone by the cube root $\sqrt[3]{}$ (one answer, no $\pm$), and $\sqrt[3]{}$ by cubing.

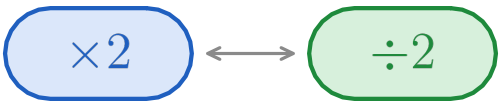
Flipping a fraction (taking the reciprocal) is undone by flipping it back.

Every operation has an **undo button**: another operation that exactly cancels it. The proper name is an **inverse operation**. Test each pair with a number: do it, then undo it, and you are back where you started.



$$7 \xrightarrow{+3} 10 \xrightarrow{-3} 7$$

plus is undone by minus (and minus by plus)



$$7 \xrightarrow{\times 2} 14 \xrightarrow{\div 2} 7$$

times is undone by divide (and divide by times)

square ⟷ $\sqrt{\ }$

$7 \xrightarrow{\text{sq}} 49 \xrightarrow{\sqrt{\ }} 7$

squaring is undone by square-rooting

flip ⟷ flip

$\frac{2}{7} \xrightarrow{\text{flip}} \frac{7}{2} \xrightarrow{\text{flip}} \frac{2}{7}$

flipping a fraction is undone by flipping it back

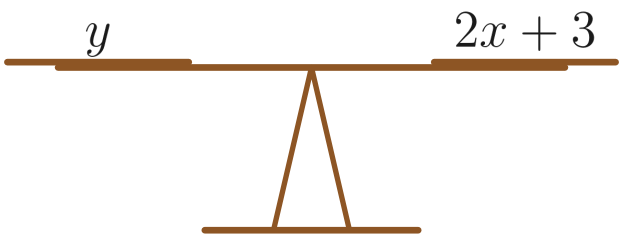
The only rule: keep the see-saw balanced

Part 0 said = is a balanced see-saw. If you take 3 off the left pan, it tips... unless you take 3 off the right pan as well. So the one and only rule of rearranging is:

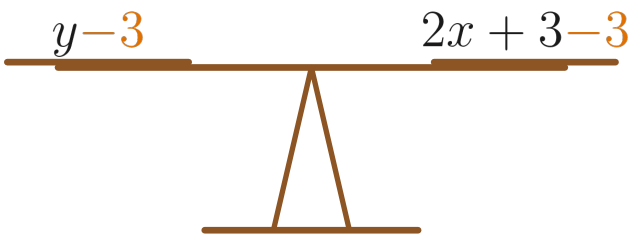
RULE: the balance rule

Whatever you do to one side, do exactly the same to the whole of the other side.

"The whole side": dividing $2x + 3$ by 2 gives $\frac{2x+3}{2}$, which is $x + \frac{3}{2}$, not $x + 3$.



balanced



take 3 from BOTH pans: still balanced

"The whole of the other side" matters. If the other side is $2x + 3$ and you divide by 2, you must divide **all** of $2x + 3$, not just the $2x$. More on this mistake later.

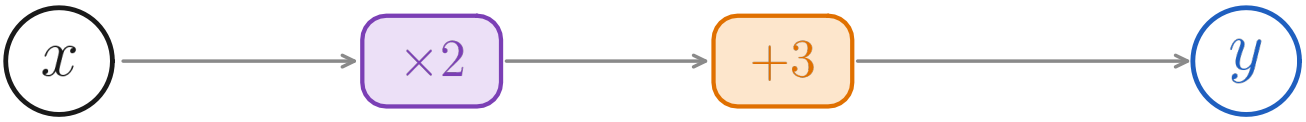
Your first unwrapping: y = 2x + 3

PROCEDURE: making a letter the subject

1. Write down what happens to the letter, **in order**, to build the formula (its layers).
2. Undo the layers in **reverse order** (last on, first off), using the undo buttons.
3. Do every move to the **whole** of both sides, one move per line.
4. Swap sides so the letter is on the left. Test with numbers.

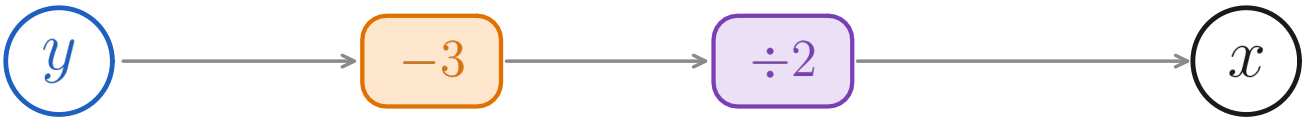
Step 1: work out how the present was wrapped. Start with x and ask: what happens to it, in order, to make y? x is first multiplied by 2 (that makes $2x$), and then 3 is added (that makes $2x + 3$).

wrapping (building the formula):

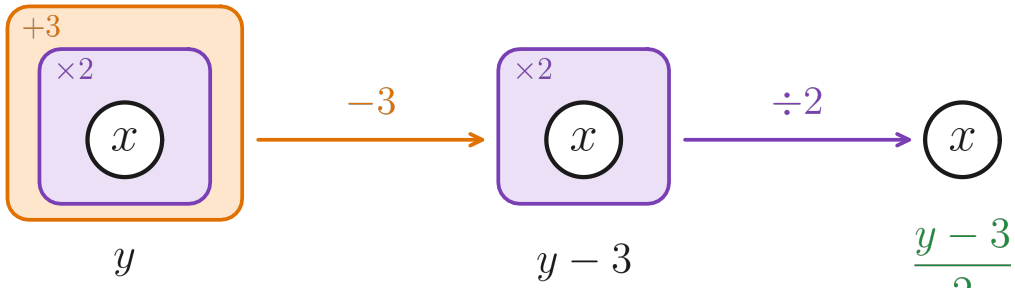


Step 2: unwrap in the reverse order, using the undo buttons. The last layer on was +3, so it comes off first, using -3. Then $\times 2$ comes off, using $\div 2$.

unwrapping (reverse order, undo buttons):



Picture it: peel the present one layer per step. Whatever each peel does to the x side, it does to the y side too:



underneath: the other side of the =

Step 3: do it on the formula, one layer per line, to both sides.

$$y = 2x + 3 \quad \longleftarrow \text{start}$$

$$y-3 = 2x + 3-3$$ ← take the outer layer off: subtract 3 from BOTH sides

$$y - 3 = 2x$$ ← +3 and -3 cancel out on the right

$$\frac{y-3}{2} = \frac{2x}{2}$$ ← next layer: divide BOTH sides by 2

$$\frac{y-3}{2} = x$$ ← 2x divided by 2 is just x

$$x = \frac{y-3}{2}$$ ← swap sides so x is on the left (allowed: = is a balance)

Test it with a number. Put $x = 4$ into the original: $y = 2 \times 4 + 3 = 11$. Now put $y = 11$ into the new formula: $x = \frac{11-3}{2} = \frac{8}{2} = 4$. We got the 4 back. The unwrapping worked. ✓

Last on, first off. Think of socks and shoes. You put socks on first and shoes second. To undress, the shoes come off first. Formulas are the same.

Why the order matters

What if you took the layers off in the wrong order, dividing by 2 first? Try it with numbers, where we know the answer should be $x = 4$ when $y = 11$:

Right order

$$11 \xrightarrow{-3} 8 \xrightarrow{\div 2} 4$$ ✓

Wrong order

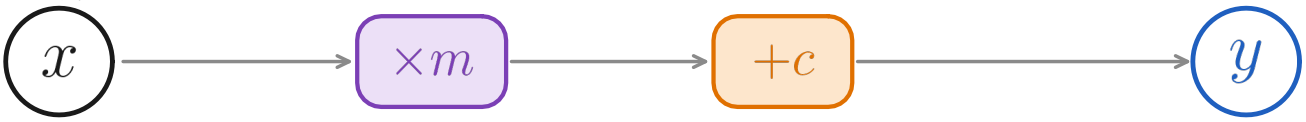
$$11 \xrightarrow{\div 2} 5.5 \xrightarrow{-3} 2.5$$ ✗

The wrong order gives 2.5, which is wrong. Undoing has to happen in the **reverse** order of doing. That is why Step 1 (working out how the formula was built) matters so much.

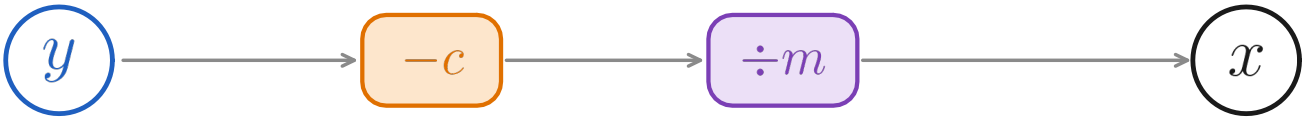
The lecture example: y = mx + c

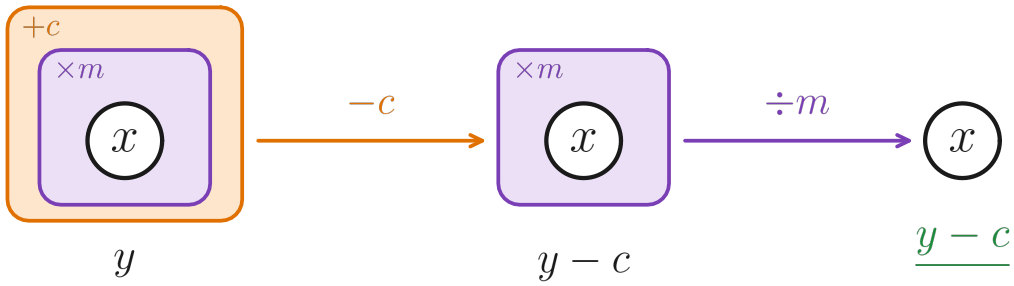
Lecture 2 rearranges the straight-line formula (you will meet it properly in Chapter 20). It is exactly the same present, with letters where the numbers were: $\times m$ instead of $\times 2$, and $+c$ instead of $+3$.

wrapping:



unwrapping:





underneath: the other side of the =

$$y - c = mx$$ ← *subtract c from both sides*

$$\frac{y - c}{m} = x$$ ← *divide both sides by m*

$$x = \frac{y - c}{m}$$ ← *swap sides*

When the letter has a minus in front

RULE: subject with a minus sign

If the letter's term is being **subtracted**, **add it to both sides** first. That moves it to the other side as a positive term.

Then unwrap as usual. (Or multiply both sides by -1, which flips every sign.)

Example: make x the subject of $y = 5 - 2x$.

$$y + 2x = 5$$ ← add 2x to both sides: now 2x is positive

$$2x = 5 - y$$ ← subtract y from both sides

$$x = \frac{5 - y}{2}$$ ← divide both sides by 2

Test: $x = 1$ gives $y = 5 - 2 = 3$. New formula: $x = \frac{5-3}{2} = 1$ ✓.

When the letter is on the bottom

RULE: subject in the denominator

Multiply both sides by the denominator first, so the letter comes up onto a top line.

Then unwrap as usual: the letter is now in an ordinary layer.

(If each side is a single fraction you may instead flip both sides; see the lecture example later.)

Example: make R the subject of $P = \frac{V}{R}$.

$$PR = V$$ ← multiply both sides by R

$$R = \frac{V}{P}$$ ← divide both sides by P

Test: $V = 12$, $R = 4$ gives $P = 3$, and $\frac{12}{3} = 4$ ✓.

Example: make v the subject of $t = \frac{d}{v+2}$. The bottom is a whole bracket:

$$t(v+2) = d$$ ← multiply both sides by (v + 2)

$$v + 2 = \frac{d}{t}$$ ← divide both sides by t

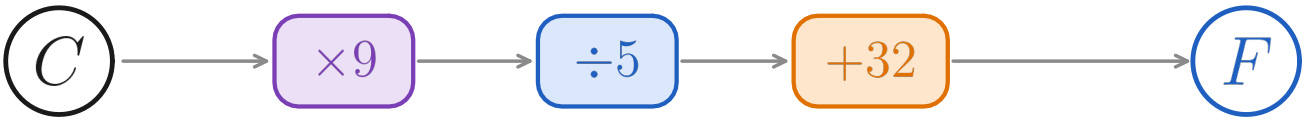
$$v = \frac{d}{t} - 2$$ ← subtract 2 from both sides

Test: $v = 3$, $d = 10$ gives $t = 2$; then $\frac{10}{2} - 2 = 3$ ✓.

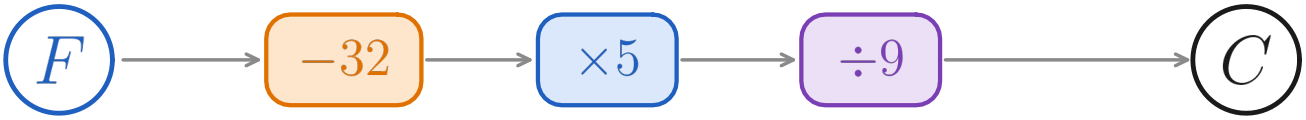
Three layers: Fahrenheit to Celsius

$F = \frac{9}{5}C + 32$ is the same as $F = \frac{9C}{5} + 32$. Make C the subject.

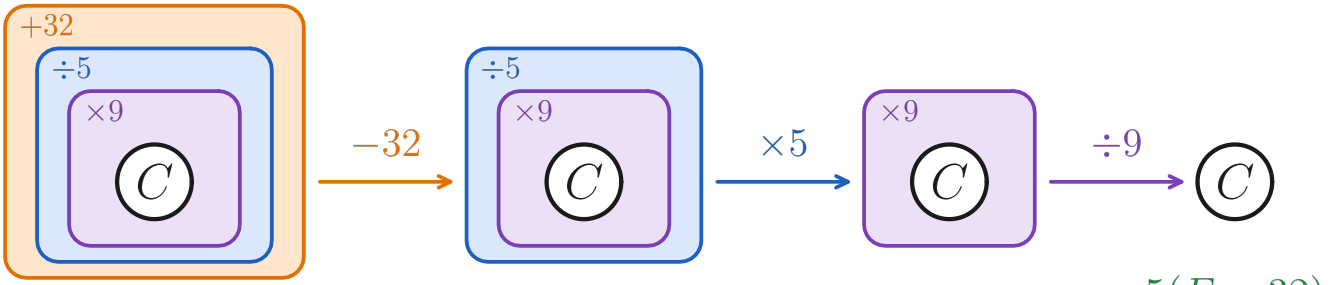
How it was wrapped: C is multiplied by 9, then divided by 5, then 32 is added.



Unwrap, last on first off: subtract 32, then **multiply** by 5 (undoes $\div 5$), then divide by 9.



Three layers, three peels:



F $\quad F - 32$ $\quad 5(F-32)$ $\quad \frac{5(F-32)}{9}$

underneath: the other side of the =

$$F = \frac{9C}{5} + 32$$ ← start

$$F - 32 = \frac{9C}{5}$$ ← outer layer: subtract 32 from both sides

$$5(F - 32) = 9C$$ ← next layer: multiply both sides by 5. Brackets: ALL of F - 32 gets x5

$$\frac{5(F - 32)}{9} = C$$ ← last layer: divide both sides by 9

$$C = \frac{5(F - 32)}{9}$$ ← swap sides

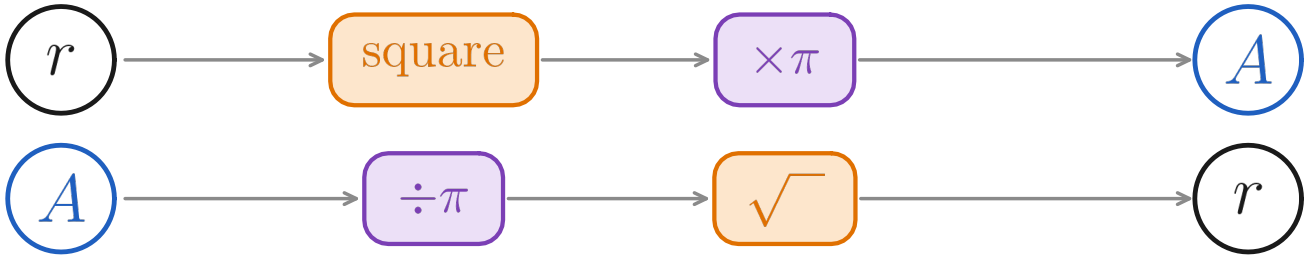
Test: water boils at 212 °F. $C = \frac{5(212-32)}{9} = \frac{5\times180}{9} = \frac{900}{9} = 100$ °C. ✓ Correct!

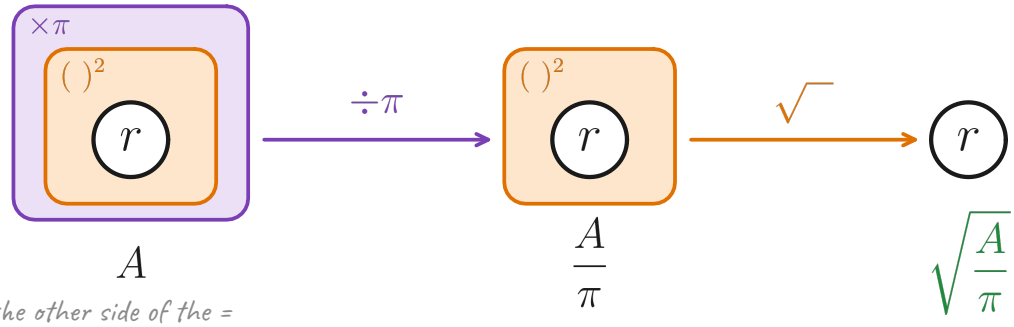
Why the bracket? Without it, $5F - 32$ would mean "multiply only F by 5, then take 32". But the whole left side, $F - 32$, has to be multiplied by 5. The bracket keeps it together.

Layers with squares and roots

The area of a circle is $A = \pi r^2$. Given the area, find the radius r.

How it was wrapped: r is squared first (that's r^2), then multiplied by π. (Squaring happens before multiplying: powers come before times in the order of operations.)





underneath: the other side of the =

$$A = \pi r^2 \quad \leftarrow \text{start}$$

$$\frac{A}{\pi} = r^2 \quad \leftarrow \text{divide both sides by pi}$$

$$\sqrt{\frac{A}{\pi}} = r \quad \leftarrow \text{square root both sides (undoes the squaring)}$$

$$r = \sqrt{\frac{A}{\pi}} \quad \leftarrow \text{swap sides}$$

Test: a circle of radius 3 has $A = \pi \times 9 \approx 28.27$. Then $\sqrt{28.27/\pi} = \sqrt{9} = 3$. ✓

A small print note: $(-3)^2$ is also 9, so strictly r could be $+3$ or -3 (written ± 3). A radius can't be negative, so here we keep the positive one. When a question isn't about a length, remember the $\pm$.

RULE: cubes and cube roots

A cube is undone by a **cube root**: if $r^3 = k$ then $r = \sqrt[3]{k}$.

There is only **one** cube root, and negatives are fine: $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$. So no $\pm$ here.

Example: the volume of a sphere is $V = \frac{4}{3}\pi r^3$. Make r the subject.

Wrapping: cube, then $\times\frac{4}{3}\pi$ (that is $\times 4$, $\div 3$, $\times \pi$). Unwrap in reverse:

$$3V = 4\pi r^3$$ ← multiply both sides by 3

$$\frac{3V}{4\pi} = r^3$$ ← divide both sides by 4 pi

$$r = \sqrt[3]{\frac{3V}{4\pi}}$$ ← cube root both sides, swap

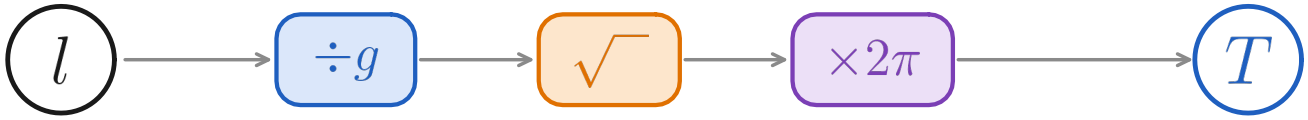
Test: $r = 2$ gives $V = \frac{4}{3}\pi \times 8 \approx 33.51$; then $\sqrt[3]{3 \times 33.51/(4\pi)} = \sqrt[3]{8} = 2$ ✓.

The pendulum, one layer at a time

This is the big example from Lecture 2 (and the one from Part 0, where you learned to read it). Make l (the string length) the subject:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

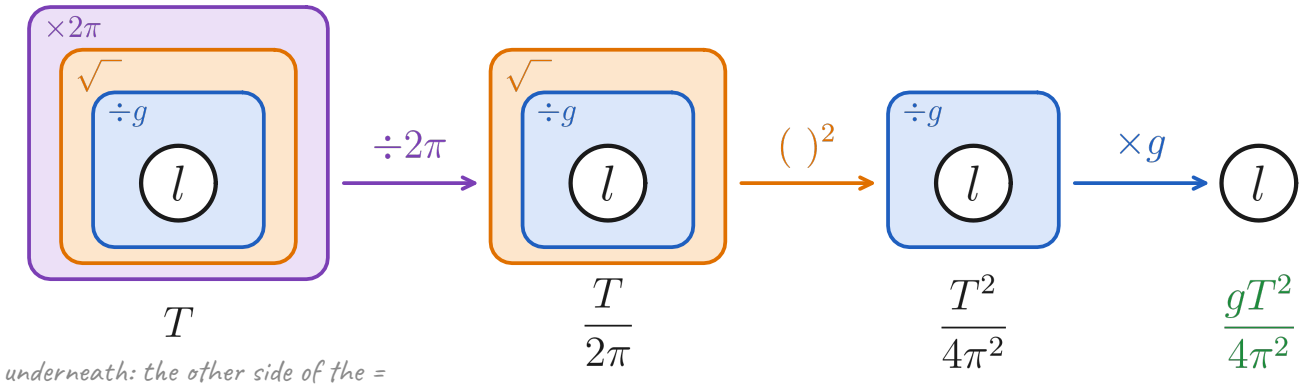
How it was wrapped, reading from the inside out: l is divided by g, then square-rooted, then multiplied by 2π.



Unwrapping (reverse order, undo buttons): divide by 2π, then square, then multiply by g.



The present, peeled from the outside in:



underneath: the other side of the =

$$T = 2\pi\sqrt{\frac{l}{g}} \quad \longleftarrow \text{start}$$

$$\frac{T}{2\pi} = \sqrt{\frac{l}{g}} \quad \longleftarrow \text{outer layer: divide both sides by 2 pi}$$

$$\left(\frac{T}{2\pi}\right)^2 = \frac{l}{g}$$

middle layer: square both sides (squaring undoes the root)

$$\frac{T^2}{4\pi^2} = \frac{l}{g}$$

tidy up: square the top (T x T) and the bottom (2 pi x 2 pi = 4 pi squared)

$$g \times \frac{T^2}{4\pi^2} = l$$

inner layer: multiply both sides by g

$$l = \frac{gT^2}{4\pi^2}$$

swap sides and tidy up. This matches your lecture slide!

Test: with $g = 9.81$ and a 1 metre string, $T = 2\pi\sqrt{1/9.81} \approx 2.006$ seconds. Put $T = 2.006$ into the new formula: $l = 9.81 \times 2.006^2/(4\pi^2) \approx 1.00$ metre. ✓

When the letter is on the bottom of a fraction

Lecture 2's hardest example: make z the subject of

$$\frac{1}{x} = \frac{2}{y} - \frac{3}{z}$$

RULE: flipping both sides

If **each side is one single fraction**, you may flip both sides: $\frac{a}{b} = \frac{c}{d}$ gives $\frac{b}{a} = \frac{d}{c}$.

Never flip term by term. Combine each side into one fraction first (Chapter 13).

This isn't a simple chain of layers, because there are several fractions. Plan: (1) get the z fraction alone on one side, (2) combine the other side into one fraction, (3) flip both sides.

$$\frac{1}{x} + \frac{3}{z} = \frac{2}{y}$$

← add 3/z to both sides (it was being taken away)

$$\frac{3}{z} = \frac{2}{y} - \frac{1}{x}$$

← subtract 1/x from both sides: now the z fraction is alone

$$\frac{3}{z} = \frac{2x}{yx} - \frac{1y}{xy}$$

← re-cut to a common bottom, xy (Chapter 13)

$$\frac{3}{z} = \frac{2x - y}{xy}$$

← same bottoms: combine the tops

$$\frac{z}{3} = \frac{xy}{2x - y}$$

← flip BOTH sides upside down. Allowed: if two things are equal, their reciprocals are equal too

$$z = \frac{3xy}{2x - y}$$ ← multiply both sides by 3

Test: pick $x = 1$, $y = 1$. The new formula gives $z = \frac{3}{2-1} = 3$. Check in the original: $\frac{1}{1} = \frac{2}{1} - \frac{3}{3}$ means $1 = 2 - 1$. ✓

Only flip when each side is ONE single fraction. Flipping $\frac{2}{y} - \frac{1}{x}$ term by term (into $\frac{y}{2} - \frac{x}{1}$) is wrong. Test with $x = 1, y = 1$: $\frac{2}{1} - \frac{1}{1} = 1$, flipped is 1; but $\frac{1}{2} - 1 = -\frac{1}{2}$. Not the same!

When the letter appears twice

PROCEDURE: subject appearing more than once

1. Clear fractions (multiply both sides by the bottom).
2. Expand brackets, then collect every term containing the letter on **one** side, everything else on the other.
3. **Factorise**: take the letter out as a common factor, so it appears once.
4. Divide both sides by the bracket (**that bracket must not be 0**).

Sometimes the letter you want is in two places, so it isn't wrapped in one neat present. Make x the subject of

$$y = \frac{x + 1}{x - 1}$$

Plan: (1) get rid of the fraction, (2) collect every x on one side, (3) take x out as a common factor, (4) divide.

$$y(x - 1) = x + 1$$ ← multiply both sides by (x - 1): the fraction disappears

$$yx - y = x + 1$$ ← open the bracket (Chapter 11)

$$yx - x = 1 + y$$ ← move the x terms left, the rest right: subtract x, add y, on both sides

$$x(y-1) = 1+y$$ ← take x out as a common factor (Chapter 12): now x appears ONCE

$$x = \frac{1+y}{y-1}$$ ← divide both sides by (y - 1)

Test: $x = 3$ gives $y = \frac{4}{2} = 2$. New formula with $y = 2$: $x = \frac{3}{1} = 3$. ✓

The factorising step is the key idea. Once all the x's are on one side, factorising turns "x in two places" into "x in one place", and then it's an ordinary one-layer unwrap.

Two safety rules

RULE: don't divide by something that could be zero

Dividing both sides by an expression is only allowed when that expression is **not 0**. In the last example, dividing by $(y-1)$ needs $y \neq 1$; write that condition down.

If you divide by a letter in an **equation**, you can lose an answer. Move everything to one side and factorise instead.

Example: $x^2 = 3x$. Dividing by x gives $x = 3$ and **loses** $x = 0$. Safe way:

$$x^2 - 3x = 0 \Rightarrow x(x-3) = 0$$ ← everything to one side, factorise

$$x = 0 \text{ or } x = 3$$ ← either factor can be 0

RULE: squaring both sides can add false answers

Squaring both sides is allowed, but it can create answers that **don't fit the original**, because 3^2 and $(-3)^2$ are both 9.

So after squaring, **check every answer in the original** and throw out any that fail.

Example: $\sqrt{x} = -3$. Squaring gives $x = 9$, but $\sqrt{9} = 3$, not -3. So there is **no solution**: a square root (the $\sqrt{}$ sign) is never negative.

Example: $\sqrt{x+3} = x - 3$. Square: $x + 3 = x^2 - 6x + 9$, so $x^2 - 7x + 6 = 0$, $(x-1)(x-6) = 0$. Check $x = 6$: $\sqrt{9} = 3 = 6 - 3$ ✓. Check $x = 1$: $\sqrt{4} = 2$ but $1 - 3 = -2$ ✗. Only $x = 6$ is an answer.

Using a rearranged formula: find the missing value

Exams rarely stop at the rearranged formula. Usually you're given **some** of the values and asked for the one that's missing:

PROCEDURE: finding an unknown from a formula

1. Circle the letter you need to find. That's the subject you want.
2. **Rearrange** the formula to make it the subject (letters only, no numbers yet).
3. **Substitute** the values you know. Put negative numbers in brackets.
4. Calculate, with units. **Check** by putting every value back into the original formula.

(Substituting first and then solving also works. Rearranging first gives a formula you can reuse, and the steps are easier to check.)

Example 1 (motion): $v^2 = u^2 + 2as$. Find s when $v = 10$, $u = 4$, $a = 3$.

$$v^2 - u^2 = 2as$$ ← rearrange: subtract u^2

$$s = \frac{v^2 - u^2}{2a}$$ ← divide by 2a, swap

$$s = \frac{10^2 - 4^2}{2 \times 3} = \frac{100 - 16}{6} = \frac{84}{6}$$ ← substitute

$$s = 14$$ ← calculate

Check: $u^2 + 2as = 16 + 2 \times 3 \times 14 = 16 + 84 = 100 = 10^2$ ✓.

Example 2 (pendulum): $T = 2\pi\sqrt{\frac{l}{g}}$. How long must the string be for $T = 2$ seconds, with $g = 9.81$?

$$l = \frac{gT^2}{4\pi^2}$$ ← rearranged earlier in this chapter

$$l = \frac{9.81 \times 2^2}{4\pi^2} = \frac{39.24}{39.48}$$ ← substitute

$$l \approx 0.994 \text{ m}$$ ← about 99 cm

Check: $2\pi\sqrt{0.994/9.81} = 2\pi \times 0.3183 \approx 2.00$ s ✓.

Example 3 (energy): $E = \frac{1}{2}mv^2$. Find v when $E = 100$ J and $m = 8$ kg.

$$v = \sqrt{\frac{2E}{m}} = \sqrt{\frac{200}{8}} = \sqrt{25}$$ ← rearrange (×2, ÷m, √), substitute

$$v = 5 \text{ m/s}$$ ← a speed: keep the + root

Check: $\frac{1}{2} \times 8 \times 5^2 = 4 \times 25 = 100$ ✓.

Three classic slip-ups (tested with numbers)

Each one starts from the same kind of formula, shows the slip, then the correct move.

Dividing only part of a side.

start: $y = 2x + 3$ — *Move: divide both sides by 2.*

wrong: ~~$\frac{y}{2} = x + 3$~~ — *only the 2x was divided; the 3 was forgotten*

right: $\frac{y}{2} = \frac{2x+3}{2} = x + \frac{3}{2}$ ✓ *the WHOLE side goes over 2, so the 3 is halved too*

Test with x = 4, y = 11: the wrong version says 5.5 = 7. The right version says 5.5 = 4 + 1.5. ✓ Divide EVERY term.

Undoing in the wrong order.

start: $y = 2x + 3$ — *Move: make x the subject.*

wrong: ~~$x = \frac{y}{2} - 3$~~ — *took off ×2 first, but it's the INNER layer*

right: $x = \frac{y-3}{2}$ ✓ *−3 first (outer layer), then ÷2*

With y = 11 the wrong one gives 2.5, the right one gives 4. ✓ Last on, first off.

Square-rooting term by term.

start: $c^2 = a^2 + b^2$ — *Move: square-root both sides to get c.*

wrong: ~~$c = a + b$~~ — *rooted a^2 and b^2 separately*

right: $c = \sqrt{a^2 + b^2}$ ✓ *the root goes over the WHOLE side*

Test with a = 3, b = 4: the right answer is the square root of 25, which is 5. The wrong one gives 7. (This is also a warning on your Lecture 2 slides.)

The unwrapping routine

1. Circle the letter you want.
2. Work out how it was wrapped: what happens to it, in order? Draw the chain.
3. Messy? Clear fractions first (multiply both sides), collect the letter on one side, and factorise so it appears once.
4. Unwrap: last layer first, using the undo buttons, doing the same to the whole of both sides.
5. Test with numbers.

Coming later: Chapter 15 uses exactly the same moves to find the value of a mystery number, and Chapter 22 adds one more undo button (the logarithm), which undoes powers like e^x. That's how Lecture 7's capacitor formula gets rearranged.

Rearranging rules

Learn these. Every worked example in the chapter is one of these rules applied step by step.

Balance rule	whatever you do to one side, do to the **whole** of the other side
Undo buttons	$+ \leftrightarrow -, \quad \times \leftrightarrow \div, \quad x^2 \leftrightarrow \pm\sqrt{x}, \quad x^3 \leftrightarrow \sqrt[3]{x}$ flip undoes flip
Order	build the chain of layers, then undo them in **reverse** order: last on, first off
Minus in front	$y = 5 - 2x \;\Rightarrow\; 2x = 5 - y$ add the subtracted term to both sides first
Letter on the bottom	$P = \dfrac{V}{R} \;\Rightarrow\; PR = V \;\Rightarrow\; R = \dfrac{V}{P}$ multiply both sides by the bottom first
Flipping	$\dfrac{a}{b} = \dfrac{c}{d} \;\Rightarrow\; \dfrac{b}{a} = \dfrac{d}{c}$ only when each side is ONE fraction
Letter twice	$yx - x = 1 + y \;\Rightarrow\; x(y-1) = 1 + y$ collect, factorise, divide
Roots	$r^2 = k \Rightarrow r = \pm\sqrt{k}, \qquad r^3 = k \Rightarrow r = \sqrt[3]{k}$ keep $+$ only for lengths, speeds and other sizes
Safety	**never divide by something that could be 0**; after squaring both sides, check every answer in the original
Find a value	rearrange first, then substitute, then check in the original formula

Your turn: fill in the chains

For each formula, fill in the wrapping boxes (what happens to the letter, in order), then the unwrapping boxes (reverse order, undo buttons), then write the rearranged formula on the line. Answers to 1 to 5 are on the last page; the later groups have theirs in grey.

1. $y = 5x - 4$ *make x the subject*

wrap: x → [ ] → [ ] → y

unwrap: y → [ ] → [ ] → x *so x =* _____

2. $P = \dfrac{V}{R}$ *make V the subject*

wrap: V → [ ] → P

unwrap: P → [ ] → V *so V =* _____

3. $v = u + at$ *make a the subject*

wrap: a → [ ] → [ ] → v

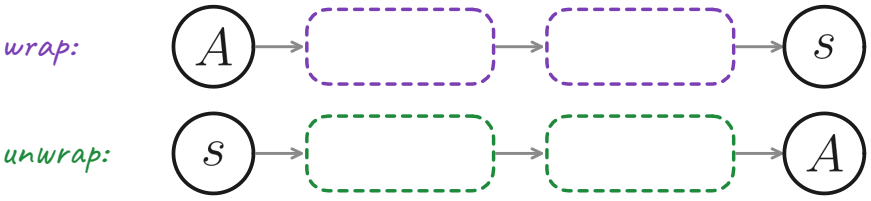
unwrap: v → [ ] → [ ] → a *so a =* _____

4. $E = \frac{1}{2}mv^2$ *make v the subject*

wrap: v → [ ] → [ ] → [ ] → E

unwrap: E → [ ] → [ ] → [ ] → v *so v =* _____

5. $s = \sqrt{\dfrac{A}{6}}$ *make A the subject*



so A = _____

Minus signs, bottoms and cubes

6. $y = 7 - 3x$: make x the subject
7. $I = V/R$: make R the subject
8. $f = 1/(x + 3)$: make x the subject
9. $V = x^3/3$: make x the subject
10. $y = (2x + 1)/(x - 3)$: make x the subject

6. $x = (7 - y)/3$ 7. $R = V/I$ 8. $x = 1/f - 3$ 9. $x = \sqrt[3]{(3V)}$ 10. $x = (3y + 1)/(y - 2)$

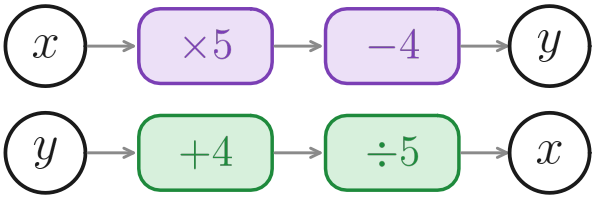
Find the unknown

11. $v^2 = u^2 + 2as$. Find s when v = 20, u = 10, a = 5.
12. $v = u + at$. Find t when v = 25, u = 5, a = 4.
13. $A = \pi r^2$. Find r when A = 154, using $\pi \approx 22/7$.
14. $F = 9C/5 + 32$. Find C when F = 50.
15. $y = (x + 1)/(x - 1)$. Find x when y = 3.
16. $E = \frac{1}{2}mv^2$. Find m when E = 90 and v = 6.

11. s = 30 ($s = (v^2 - u^2)/(2a) = 300/10$) 12. t = 5 ($t = (v - u)/a$) 13. r = 7 ($r^2 = A/\pi = 49$)
14. C = 10 ($C = 5(F - 32)/9$) 15. x = 2 ($x = (y + 1)/(y - 1)$) 16. m = 5 ($m = 2E/v^2 = 180/36$)

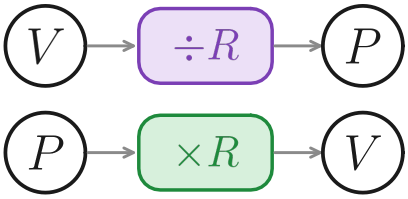
Answers

1.



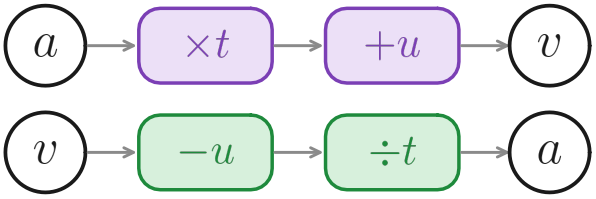
$$x = \frac{y+4}{5}$$

2.



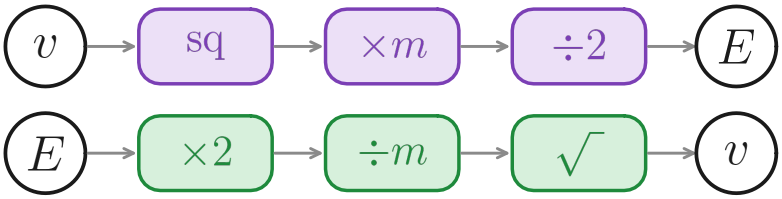
$$V = PR$$

3.



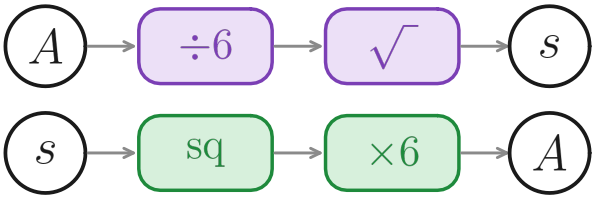
$$a = \frac{v-u}{t}$$

4.



$$v = \sqrt{\frac{2E}{m}}$$

5.



$$A = 6s^2$$

In question 4, $\frac{1}{2}mv^2$ means: square v, times by m, then halve (divide by 2). So unwrapping is: double, divide by m, square root.